



Research paper

Closed-form forward kinematics, singularity, dexterity and workspace analyses of a special class of 3-RRR spherical parallel mechanisms

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ABSTRACT

3-RRR spherical parallel mechanisms (SPMs) are capable of producing three-DOF spherical motion, which can find applications in various fields. This paper investigates the kinematics and workspace performance of a particular class of 3-RRR SPMs, characterized by coaxial input-joint axes, coplanar distal-joint axes, and six orthogonal links. First, two novel formulations are proposed for the forward kinematics (FK) of this robot class, both resulting in linear univariate polynomials that significantly streamline the FK formulation and reduce the computational complexity. Moreover, this robot class exhibits minimal and physically interpretable singularity loci in both the orientation workspace and the input joint space, greatly facilitating singularity avoidance and path planning. Furthermore, local and global dexterity indices are employed to analyze the mechanism's performance, from which the architectural parameters yielding optimal dexterity are identified. Finally, the robot class demonstrates a considerably large tilt-torsion orientation workspace when the architectural parameters and link shapes are appropriately selected, allowing for infinite torsional motion when pointed within almost a hemisphere. The foregoing features make the robot class potentially promising in a wide range of applications.

1. Introduction

Three-degree-of-freedom (three-DOF) spherical parallel mechanisms (SPMs) have attracted considerable research interest over the years due to their broad applicability in areas such as robotic shoulders and wrists, orienting devices, haptic interfaces, rehabilitation devices, and spherical motors [1–5]. Among the various proposed three-DOF SPM topologies, 3-RRR SPMs have been extensively studied, with numerous investigations pertaining to their forward kinematics (FK) [6–16], singularity [17–19], dexterity [20,21], workspace [22–24], and design optimization [25,26].

As is well known, the FK univariate polynomial of general 3-RRR SPMs is octic [6,7], and considerable efforts have been devoted to identifying those admitting closed-form FK solutions (CFFKS) by exploiting special geometries such as orthogonality, coplanarity, and coaxiality [9–15]. Such mechanisms bear simple FK relations that can be solved efficiently and robustly even when operating near singular postures, making them particularly favorable from the kinematic perspective [11,14]. In several recent works of the

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authors [13–15], four broad classes of 3-RRR SPMs with CFFKS have been identified, for which FK univariate polynomials of linear, quadratic, or quartic degree were obtained. Notably, one of these classes — characterized by coplanar intermediate-joint axes and coplanar distal-joint axes — was found to possess only four assembly modes, as opposed to the general case of eight [15]. Moreover, for a special subclass within this class—defined by the additional condition of three orthogonal distal links, quadratic univariate polynomials have been obtained for its FK problem, further simplifying the formulation [15].

In this paper, the FK of the aforementioned subclass — characterized by coaxial input-joint axes, coplanar distal-joint axes, and six orthogonal links — is revisited, and two novel formulations are proposed, both yielding linear univariate polynomials that further streamline the FK formulation and reduce computational cost. Moreover, the singularity, dexterity, and workspace characteristics of this subclass are investigated, revealing several distinct and attractive features, such as minimal singularity loci that are both intuitive and physically interpretable, as well as a considerably large tilt-torsion orientation workspace that enables infinite torsional motion when pointed within almost a hemisphere. These properties make the robot class under study promising for various applications, such as orienting devices, robotic wrists and shoulders that require continuous torsional motion across a wide range of directions [1,2].

Singularity analysis is fundamental to the design, trajectory planning, and control of parallel mechanisms (PMs), as singular configurations can severely compromise a system's mobility, controllability, and structural integrity [27,28]. For 3-RRR SPMs, their singularities are found to be typically associated with complex two-dimensional surfaces embedded within the three-dimensional orientation workspace, which not only complicate path planning but also significantly limit the usable workspace [17,18]. Interestingly, the 3-RRR robot class under study exhibits remarkably simple singularity loci that are both intuitive and physically interpretable. As will be detailed in Section 4, this robot class encounters singularities if and only if (iff) the normal vector of the moving platform (MP) becomes aligned with or perpendicular to the coaxial input-joint axes—a strikingly simple geometric condition that not only facilitates singularity avoidance but also significantly enlarges the usable orientation workspace.

Dexterity is another critical performance metric in the evaluation of PMs, as it affects the control accuracy and the directional homogeneity of the MP motion [29–31]. In this paper, the dexterity of the proposed 3-RRR robot class is analyzed based on the condition number of its Jacobian matrix [32,33], with particular attention to the distribution of the local dexterity index within the orientation workspace, as well as the influence of its architectural parameters on the global dexterity index. These results provide practical guidelines for selecting architecture parameters to optimize the robot's operational performance from the perspective of dexterity.

Workspace analysis is also an essential component in the design of PMs, and is commonly performed using geometrical [34,35], discretization [36,37], or numerical [38,39] methods. Among these, the discretization method is the most widely used due to its effectiveness in handling various constraints such as singularities, mechanical interference, and joint limits, as adopted in this paper. Although PMs generally exhibit limited workspace due to their multi-loop architecture and complex singularity distribution [40,41], the proposed 3-RRR robot class demonstrates a considerably large tilt-torsion orientation workspace, enabling infinite torsional motion when pointed within almost a hemisphere and outperforming most previously reported SPMs in this regard [42–44], making it a promising candidate for various applications.

The main contributions of the paper are summarized as follows:

1. A special 3-RRR SPM robot class is investigated, characterized by a combination of geometric requirements that gives rise to various distinctive and attractive features.
2. Two simplified FK formulations yielding linear univariate polynomials are proposed, which, to the best of authors' knowledge, make the considered robot class the third known subclass of 3-RRR SPMs that admits linear univariate polynomials [12–14].
3. Explicit analytic singularity conditions are derived in both the MP orientation workspace and the input joint space, greatly facilitating the identification of singular configurations. Notably, the singularity loci in the orientation workspace are identical across the robot class, which are minimal, intuitive, and physically interpretable.
4. The dexterity performance of the robot class is evaluated using both local and global dexterity indices, providing guidance on parameter selection to enhance the robot's operational performance from the dexterity perspective.
5. The workspace performance of the robot class is evaluated, showing that, upon proper design of the relevant link shapes, it admits unlimited torsional motion over a wide range of MP normal vector directions, covering nearly a hemisphere.

The remainder of the paper is organized as follows: Section 2 introduces the architecture of the robot class under study, along with the relevant notations. Section 3 presents two different formulations for its FK, accompanied by several remarks and an illustrative case study to validate the proposed formulations. Moreover, Sections 4–6 are devoted to the singularity, dexterity and workspace analyses, respectively, revealing several attractive features of the robot class under study. Section 7 concludes the paper.

2. Architecture description

An example of the 3-RRR robot class under study is illustrated in Fig. 1(a), which comprises an MP, a base platform (BP), and three identical limbs formed by three serially connected revolute (R) joints. Now, the three R joints in each limb, ordered from the BP to the MP, are termed the proximal, intermediate and distal joints, with the unit direction vectors parallel to their axes denoted as $\mathbf{u}_i$, $\mathbf{v}_i$ and $\mathbf{w}_i$, respectively, for $i = 1, 2, 3$, as shown in Fig. 1(b). Moreover, these axes are arranged as follows:

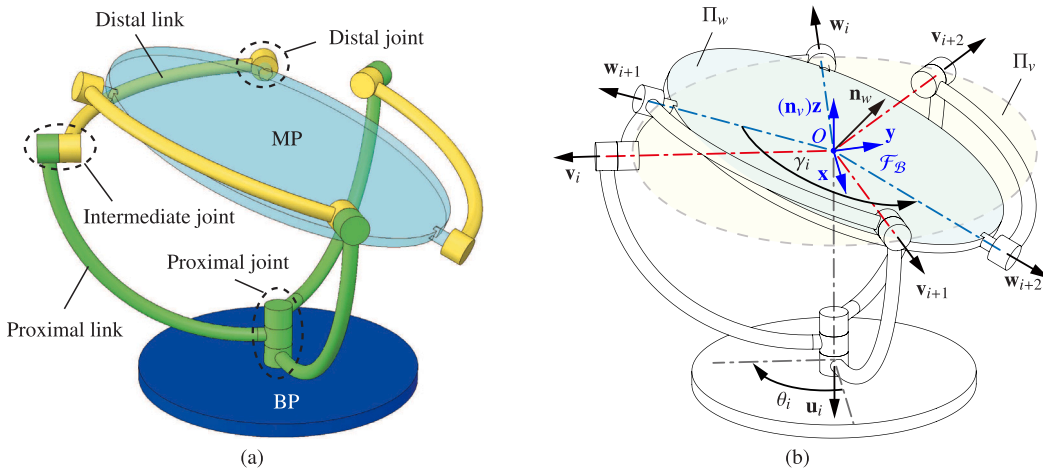


Fig. 1. (a) An example of the 3-RRR robot class under study in an arbitrary posture, and (b) its corresponding notation.

- (1) The three distal-joint axes are coplanar yet mutually non-collinear.¹
- (2) The three proximal-joint axes are coaxially arranged, i.e., $\mathbf{u}_1 = \mathbf{u}_2 = \mathbf{u}_3$.
- (3) For each limb, the intermediate-joint axis is perpendicular to both the proximal-joint and distal-joint axes, i.e., $\mathbf{v}_i \perp \mathbf{u}_i$ and $\mathbf{v}_i \perp \mathbf{w}_i$, for $i = 1, 2, 3$.

Interestingly, the first condition implies that $\{\mathbf{w}_i\}_1^3$ lie in a common plane, whereas the last two conditions imply that $\{\mathbf{v}_i\}_1^3$ also lie in a common plane. Without loss of generality, we denote these planes as Π_w and Π_v , respectively, with their corresponding unit normal vectors $\mathbf{n}_w$ and $\mathbf{n}_v$ oriented as shown in Fig. 1(b). It is noteworthy that, despite the aforementioned geometric constraints, the 3-RRR robot class still encompasses a relatively large set of designs, as the angles between the three coplanar vectors $\{\mathbf{w}_i\}_1^3$ on the MP can be varied arbitrarily, provided they remain non-collinear. To describe the MP architecture, we further introduce γ_i to represent the angle from $\mathbf{w}_{i+1}$ to $\mathbf{w}_{i+2}$, measured about $\mathbf{n}_w$, for $i = 1, 2, 3$.²

Next, in order to facilitate the description of the input joint variables, a base frame $\mathcal{F}_B$ is established, with its origin located at the center of rotation O of the MP, its z -axis aligned with $\mathbf{n}_v$ and the x -axis arbitrarily chosen within the plane normal to the z -axis, as shown in Fig. 1(b). Accordingly, the set of input joint variables is denoted as $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^T$, with θ_i defined as the angle from the x -axis to $\mathbf{v}_i$, measured about $\mathbf{u}_i$.

Finally, it is worth noting that the coaxial arrangement of the three input revolute joints — adopted in several 3-RRR SPMs [15–17,43–52] — endows these mechanisms with infinite torsional capability about certain directions, a particularly attractive feature for applications such as orienting devices, robotic wrists and shoulders [1,2]. As will be shown later, the robot class under study can achieve a large tilt-torsion orientation workspace, enabling infinite torsional motion when pointed within almost a hemisphere. Moreover, the three coaxial input R joints can be driven by three sets of gear or belt transmission systems [44–47], or alternatively, implemented as three sliders moving on circular rails with a common axis—the latter being beneficial for reducing space requirements along the common axis. See Refs. [17,42,53,54] for several examples of such implementations.

3. Forward kinematics of the robot class under study

As mentioned in Section 1, the FK of the robot class under study has been previously investigated in a prior work of the authors, where quadratic univariate polynomials were obtained [14,15]. Interestingly, it has been recently discovered that for this robot class, the robot postures corresponding to all of the FK solutions for a given set of input joint variables result in the same line of intersection between planes Π_v and Π_w —namely, the planes containing the coplanar intermediate-joint axes and the coplanar distal-joint axes, respectively. Moreover, the angle between the normal vectors of these two planes remains constant across these postures. Based on this observation, two novel formulations are proposed, both yielding linear univariate polynomials that further streamline the FK process and reduce computational cost.

¹ The case where two of $\{\mathbf{w}_i\}_1^3$ are parallel is not considered, because such an MP layout — when applied to the robot class under study — renders the mechanism permanently singular, as will be shown in Section 4.1.2, making such configurations impractical.

² Subscripts $i+1$ and $i+2$, when exceeding 3, are taken as $i-2$ and $i-1$, respectively.

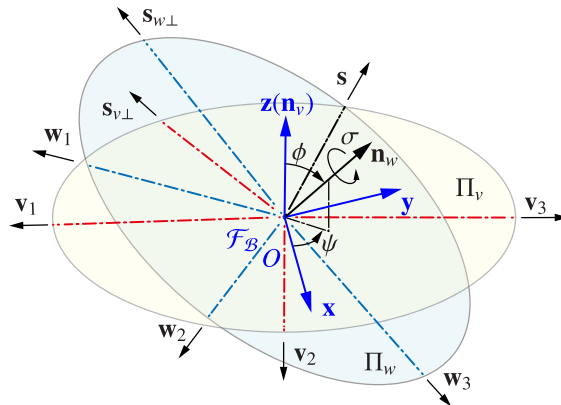


Fig. 2. Illustration of the azimuth-tilt-torsion angles and the directions of unit vectors relevant to the FK formulations, with the yellow and blue planes representing Π_v and Π_w , respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.1. Two formulations yielding linear univariate polynomials

Before describing the first formulation, we first define the directions of $\{\mathbf{w}_i\}_1^3$ under the reference posture as

$$[\mathbf{w}_{10}]_B = [1, 0, 0]^T, \quad [\mathbf{w}_{20}]_B = [\cos \gamma_3, \sin \gamma_3, 0]^T, \quad [\mathbf{w}_{30}]_B = [\cos \gamma_2, -\sin \gamma_2, 0]^T \quad (1)$$

where the subscript B indicates that the vector is expressed in the base frame. Moreover, we employ the azimuth-tilt-torsion (ψ - ϕ - σ) angle convention [17] to represent the orientation of the MP in this paper, where $\psi \in [0, 2\pi)$, $\phi \in [0, \pi]$, and $\sigma \in [0, 2\pi)$, as illustrated in Fig. 2. Then, the rotation matrix $\mathbf{Q}$ of the MP can be expressed as

$$\begin{aligned} \mathbf{Q}(\psi, \phi, \sigma) &= \mathbf{R}_z(\psi) \mathbf{R}_y(\phi) \mathbf{R}_x(\sigma - \psi) \\ &= \begin{bmatrix} c(\psi)c(\phi)c(\sigma - \psi) - s(\psi)s(\sigma - \psi) & -c(\psi)c(\phi)s(\sigma - \psi) - s(\psi)c(\sigma - \psi) & c(\psi)s(\phi) \\ s(\psi)c(\phi)c(\sigma - \psi) + c(\psi)s(\sigma - \psi) & -s(\psi)c(\phi)s(\sigma - \psi) + c(\psi)c(\sigma - \psi) & s(\psi)s(\phi) \\ -s(\phi)c(\sigma - \psi) & s(\phi)s(\sigma - \psi) & c(\phi) \end{bmatrix} \end{aligned} \quad (2)$$

where $c(\cdot)$ and $s(\cdot)$ represent the cosine and sine functions, respectively. Then, the unit normal vectors of planes Π_v and Π_w can be respectively expressed as

$$[\mathbf{n}_v]_B = [0, 0, 1]^T, \quad [\mathbf{n}_w]_B = [\cos \psi \sin \phi, \sin \psi \sin \phi, \cos \phi]^T \quad (3)$$

Next, we introduce $\mathbf{s}$ as the unit vector parallel to the line of intersection between planes Π_v and Π_w , which can be calculated as

$$[\mathbf{s}]_B = (\mathbf{n}_v \times \mathbf{n}_w) / \|\mathbf{n}_v \times \mathbf{n}_w\|_2 = [-\sin \psi, \cos \psi, 0]^T \quad (4)$$

Moreover, $\mathbf{s}_{v\perp}$ and $\mathbf{s}_{w\perp}$ represent two unit vectors that are orthogonal to $\mathbf{s}$, lying within planes Π_v and Π_w , respectively, whose expressions can be determined as

$$\begin{aligned} [\mathbf{s}_{v\perp}]_B &= \mathbf{n}_v \times \mathbf{s} = [-\cos \psi, -\sin \psi, 0]^T \\ [\mathbf{s}_{w\perp}]_B &= \mathbf{n}_w \times \mathbf{s} = [-\cos \psi \cos \phi, -\sin \psi \cos \phi, \sin \phi]^T \end{aligned} \quad (5)$$

Furthermore, since $\{\mathbf{v}_i\}_1^3$ and $\{\mathbf{w}_i\}_1^3$ lie in planes Π_v and Π_w , respectively, they can be decomposed accordingly as

$$\mathbf{v}_i = \cos \vartheta_{vi} \mathbf{s} + \sin \vartheta_{vi} \mathbf{s}_{v\perp}, \quad \mathbf{w}_i = \cos \vartheta_{wi} \mathbf{s} + \sin \vartheta_{wi} \mathbf{s}_{w\perp} \quad (6)$$

where ϑ_{vi} denotes the angle from $\mathbf{s}$ to $\mathbf{v}_i$ (measured about $\mathbf{n}_v$), and ϑ_{wi} denotes the angle from $\mathbf{s}$ to $\mathbf{w}_i$ (measured about $\mathbf{n}_w$), for $i = 1, 2, 3$.

Now, three constraint equations can be formulated based on the orthogonality conditions between each pair of $\{\mathbf{v}_i, \mathbf{w}_i\}_1^3$, namely,

$$\mathbf{v}_i \cdot \mathbf{w}_i = 0, \quad \text{for } i = 1, 2, 3 \quad (7)$$

which, after inserting Eqs. (4)–(6) in, becomes

$$\cos \phi \sin \vartheta_{wi} \sin \vartheta_{vi} + \cos \vartheta_{wi} \cos \vartheta_{vi} = 0, \quad \text{for } i = 1, 2, 3 \quad (8)$$

Next, we proceed to eliminate ϕ upon rewriting Eq. (8) into the following form:

$$\cos \phi = -\cot \vartheta_{wi} \cot \vartheta_{vi}, \quad \text{for } i = 1, 2, 3 \quad (9)$$

from which two constraint equations independent of ϕ can be readily established, namely,

$$\begin{aligned} \cot \vartheta_{w1} \cot \vartheta_{v1} &= \cot \vartheta_{w2} \cot \vartheta_{v2} \\ \cot \vartheta_{w1} \cot \vartheta_{v1} &= \cot \vartheta_{w3} \cot \vartheta_{v3} \end{aligned} \quad (10)$$

where $\{\vartheta_{vi}\}_1^3$ and $\{\vartheta_{wi}\}_1^3$ appear to be all unknowns. However, ϑ_{v2} , ϑ_{v3} , ϑ_{w2} and ϑ_{w3} can be eliminated simply upon resorting to the following relationship:

$$\vartheta_{v2} = \vartheta_{v1} + \nu_{12}, \quad \vartheta_{v3} = \vartheta_{v1} + \nu_{13}, \quad \vartheta_{w2} = \vartheta_{w1} + \omega_{12}, \quad \vartheta_{w3} = \vartheta_{w1} + \omega_{13} \quad (11)$$

where $\nu_{ij} \equiv \theta_i - \theta_j$, $\omega_{12} \equiv \gamma_3$ and $\omega_{13} \equiv -\gamma_2$ are all available in the FK problem. After substituting Eq. (11) into Eq. (10), a system of two equations in ϑ_{v1} and ϑ_{w1} is obtained, namely,

$$\begin{aligned} p_1 \cot \vartheta_{v1} \cot^2 \vartheta_{w1} + q_1 (\cot^2 \vartheta_{v1} + 1) \cot \vartheta_{w1} + r_1 &= 0 \\ p_2 \cot \vartheta_{v1} \cot^2 \vartheta_{w1} + q_2 (\cot^2 \vartheta_{v1} + 1) \cot \vartheta_{w1} + r_2 &= 0 \end{aligned} \quad (12)$$

with

$$\begin{aligned} p_1 &\equiv (\cot \vartheta_{v1} + \cot \nu_{12}), & q_1 &\equiv \cot \omega_{12}, & r_1 &\equiv \cot \nu_{12} \cot \vartheta_{v1} - 1 \\ p_2 &\equiv (\cot \vartheta_{v1} + \cot \nu_{13}), & q_2 &\equiv \cot \omega_{13}, & r_2 &\equiv \cot \nu_{13} \cot \vartheta_{v1} - 1 \end{aligned} \quad (13)$$

Then, taking p_2 times the first equation of Eq. (12) minus p_1 times the second, or q_2 times the first equation of Eq. (12) minus q_1 times the second, we obtain

$$A_1 (\cot^2 \vartheta_{v1} + 1) \cot \vartheta_{w1} = B_1 (\cot^2 \vartheta_{v1} + 1), \quad A_1 \cot \vartheta_{v1} \cot^2 \vartheta_{w1} = B_2 \quad (14)$$

or, equivalently,

$$A_1 \cot \vartheta_{w1} = B_1, \quad (A_1 \cot \vartheta_{w1})^2 \cot \vartheta_{v1} = A_1 B_2 \quad (15)$$

with

$$\begin{aligned} A_1 &= (\cot \omega_{13} - \cot \omega_{12}) \cot \vartheta_{v1} - \cot \omega_{12} \cot \nu_{13} + \cot \omega_{13} \cot \nu_{12}, \\ B_1 &= \cot \nu_{12} - \cot \nu_{13}, \quad B_2 = \cot \omega_{13} - \cot \omega_{12} + (\cot \omega_{12} \cot \nu_{13} - \cot \omega_{13} \cot \nu_{12}) \cot \vartheta_{v1} \end{aligned} \quad (16)$$

Then, $\cot \vartheta_{w1}$ can be readily eliminated upon substituting the first of Eq. (15) into the second, leading to a quadratic univariate polynomial in $\cot \vartheta_{v1}$, namely,

$$\zeta_2 \cot^2 \vartheta_{v1} + \zeta_1 \cot \vartheta_{v1} - \zeta_2 = 0 \quad (17)$$

with

$$\begin{aligned} \zeta_2 &= (\cot \omega_{12} - \cot \omega_{13})(\cot \omega_{12} \cot \nu_{13} - \cot \omega_{13} \cot \nu_{12}) \\ \zeta_1 &= (\cot \omega_{12} \cot \nu_{13} - \cot \omega_{13} \cot \nu_{12})^2 + (\cot \nu_{12} - \cot \nu_{13})^2 - (\cot \omega_{12} - \cot \omega_{13})^2 \end{aligned} \quad (18)$$

Interestingly, it can be observed that the coefficient of the quadratic term in Eq. (17) is equal to the negative of its constant term, thereby enabling Eq. (17) to be rewritten as a linear univariate polynomial in $\cot(2\vartheta_{v1})$ via the double-angle identity [55], namely,

$$\cot(2\vartheta_{v1}) = \frac{\cot^2 \vartheta_{v1} - 1}{2 \cot \vartheta_{v1}} = -\frac{\zeta_1}{2\zeta_2} \quad (19)$$

This implies that a unique value of $\cot(2\vartheta_{v1})$ can be determined for a given set of input joint variables, from which the MP orientation can then be readily obtained, as will be detailed in Section 3.2.

Next, we present an alternative formulation for the FK problem of the robot class under study, which differs from the previous one by addressing Eq. (12) in a different manner. To achieve this, we expand Eq. (12) as

$$\begin{aligned} \cot^2 \vartheta_{v1} \cot^2 \vartheta_{w1} + \cot \omega_{12} \cot^2 \vartheta_{v1} \cot \vartheta_{w1} + \cot \nu_{12} \cot \vartheta_{v1} \cot^2 \vartheta_{w1} + \cot \nu_{12} \cot \vartheta_{v1} + \cot \omega_{12} \cot \vartheta_{w1} &= 1 \\ \cot^2 \vartheta_{v1} \cot^2 \vartheta_{w1} + \cot \omega_{13} \cot^2 \vartheta_{v1} \cot \vartheta_{w1} + \cot \nu_{13} \cot \vartheta_{v1} \cot^2 \vartheta_{w1} + \cot \nu_{13} \cot \vartheta_{v1} + \cot \omega_{13} \cot \vartheta_{w1} &= 1 \end{aligned} \quad (20)$$

which, after multiplying their both sides by $\sin^2 \vartheta_{v1} \sin^2 \vartheta_{w1}$ and some manipulations, can be simplified to

$$\begin{aligned} \cos(2\vartheta_{v1}) + \cot \nu_{12} \sin(2\vartheta_{v1}) + \cot \omega_{12} \sin(2\vartheta_{w1}) + \cos(2\vartheta_{w1}) &= 0 \\ \cos(2\vartheta_{v1}) + \cot \nu_{13} \sin(2\vartheta_{v1}) + \cot \omega_{13} \sin(2\vartheta_{w1}) + \cos(2\vartheta_{w1}) &= 0 \end{aligned} \quad (21)$$

Then, by subtracting the second equation from the first, or by taking $\cot \omega_{13}$ times the first of Eq. (21) minus $\cot \omega_{12}$ times the second, Eq. (21) can be reorganized into the following form:

$$\begin{aligned} \sin(2\vartheta_{w1})(\cot \omega_{12} - \cot \omega_{13}) &= -(\cot \nu_{12} - \cot \nu_{13}) \sin(2\vartheta_{v1}) \\ \cos(2\vartheta_{w1})(\cot \omega_{12} - \cot \omega_{13}) &= (\cot \nu_{12} \cot \omega_{13} - \cot \nu_{13} \cot \omega_{12}) \sin(2\vartheta_{v1}) - (\cot \omega_{12} - \cot \omega_{13}) \cos(2\vartheta_{v1}) \end{aligned} \quad (22)$$

where $\cos(2\vartheta_{w1})$ and $\sin(2\vartheta_{w1})$ can be eliminated upon squaring both sides of the two equations and then adding them up. This yields a univariate polynomial in terms of ϑ_{v1} , namely,

$$E \cos(2\vartheta_{v1}) \sin(2\vartheta_{v1}) - F \sin^2(2\vartheta_{v1}) = 0 \quad (23)$$

with

$$\begin{aligned} E &= 2(\cot \omega_{12} - \cot \omega_{13})(\cot \omega_{12} \cot v_{13} - \cot \omega_{13} \cot v_{12}) \\ F &= -(\cot \omega_{12} \cot v_{13} - \cot \omega_{13} \cot v_{12})^2 - (\cot v_{12} - \cot v_{13})^2 + (\cot \omega_{12} - \cot \omega_{13})^2 \end{aligned} \quad (24)$$

To this end, a linear univariate polynomial in $\cot(2\vartheta_{v1})$ can be obtained upon dividing both sides of Eq. (23) by $\sin^2(2\vartheta_{v1})$, namely,

$$\cot(2\vartheta_{v1}) = F/E \quad (25)$$

Interestingly, the symbolic expressions for $\cot(2\vartheta_{v1})$, as displayed in Eqs. (19) and (25), are exactly the same.

It is worth mentioning that, to the best of the authors' knowledge, the robot class under study represents the third known subclass of 3-RRR SPMs admitting linear univariate polynomials [12–14], despite the extensive research on this topic. Moreover, the geometric requirements defining the robot class, as introduced in Section 2, are fundamental for these simplified FK formulations. Relaxing any of these requirements typically leads to more involved FK formulations; for instance, removing the requirement of three orthogonal distal links yields a quartic polynomial at best [15], while further removing the requirement of three orthogonal proximal links results in an octic polynomial [6].

3.2. The determination of the MP orientation

Next, we proceed with determining the MP orientation based on the value of $\cot(2\vartheta_{v1})$. Since a unique value of $\cot(2\vartheta_{v1})$ can be calculated based on Eqs. (19) or (25), up to four distinct values of ϑ_{v1} can be obtained, namely,

$$\vartheta_{v1} = \vartheta_{v10} + k\pi/2, \quad \text{with } \vartheta_{v10} = \text{arccot}(F/E)/2 \in (0, \pi/2) \quad \text{and } k = 0, 1, 2, 3 \quad (26)$$

Moreover, for each ϑ_{v1} thus obtained, two possible values of ϑ_{w1} can be calculated via Eqs. (15) or (22), resulting in up to eight distinct pairs of $\{\vartheta_{v1}, \vartheta_{w1}\}$. Furthermore, for each pair of $\{\vartheta_{v1}, \vartheta_{w1}\}$, a unique set of ϑ_{v2} , ϑ_{v3} , ϑ_{w2} , and ϑ_{w3} can be determined using Eq. (11), while a unique value of $\cos \phi$ can be computed via the relation $\cos \phi = -\cot \vartheta_{w1} \cot \vartheta_{v1}$, leading to a unique value of ϕ within the range of $[0, \pi]$. Next, upon inserting Eqs. (4) and (5) into the first of Eq. (6), we obtain

$$[\mathbf{v}_1]_B = [-\sin(\vartheta_{v1} + \psi), \cos(\vartheta_{v1} + \psi), 0]^T \equiv [v_{1x}, v_{1y}, 0]^T \quad (27)$$

Since v_{1x} and v_{1y} are already available in the FK problem, and ϑ_{v1} has been obtained in the previous steps, a unique value of ψ can now be determined from Eq. (27) for each ϑ_{v1} thus obtained. To this end, ϕ , ψ , $\{\vartheta_{vi}\}_1^3$ and $\{\vartheta_{wi}\}_1^3$ are all available, from which $\{\mathbf{w}_i\}_1^3$ and the normal vector $\mathbf{n}_w$ of the MP can be readily calculated using Eqs. (3)–(6). Lastly, the torsion angle σ can be determined as the rotation angle of $\mathbf{w}_1$ from its zero-torsion direction $\mathbf{w}_{1z}$ (with ϕ , ψ specified as their actual values) to its actual direction, measured about $\mathbf{n}_w$. Accordingly, it can be calculated as

$$\sigma = \text{sgn}(\mathbf{w}_{1z} \times \mathbf{w}_1 \cdot \mathbf{n}_w) \arccos(\mathbf{w}_{1z} \cdot \mathbf{w}_1) \quad (28)$$

where $\text{sgn}(\cdot)$ denotes the signum function, and

$$[\mathbf{w}_{1z}]_B = \mathbf{Q}(\psi, \phi, 0)[\mathbf{w}_{10}]_B = [\cos \phi \cos^2 \psi + \sin^2 \psi, (\cos \phi - 1) \cos \psi \sin \psi, -\cos \psi \sin \phi]^T \quad (29)$$

Now, all three orientation parameters of the MP are determined, thus resolving the orientation problem.

It is noteworthy that, although the above procedure appears to produce up to eight distinct FK solutions, at most four of these solutions are valid, which can be identified by the condition $|\cos \phi| \leq 1$. Indeed, in our prior work [15], the FK problem of a broader class of 3-RRR SPMs was investigated, characterized by three coplanar intermediate-joint axes and three coplanar distal-joint axes. Moreover, quartic univariate polynomials were obtained for that broader class, based on which a rigorous proof was provided showing that any 3-RRR SPM within that broader class admits at most four assembly modes. Since the robot class studied here is a subclass of this broader class, it directly follows that it also admits at most four assembly modes.

3.3. Remarks on the proposed formulations

Next, we consider the special cases in the aforementioned FK formulation that may require particular attention. Specifically, the proposed FK formulations may encounter cotangent terms approaching infinity — arising from $\{\cot \omega_{ij}\}_2^3$, $\{\cot v_{ij}\}_2^3$, $\{\cot \vartheta_{vi}\}_1^3$ and $\{\cot \vartheta_{wi}\}_1^3$ — or vanishing denominators, which appear in Eqs. (19) or (25). These cases are discussed as follows:

- (1) $\{\cot \omega_{ij}\}_2^3$ will never approach infinity, since the three distal-joint axes of the MP are never parallel, as explained in footnote (1).
- (2) $\{\cot v_{ij}\}_2^3$ will approach infinity when the intermediate-joint axis of the second or third limb becomes parallel to that of the first limb, which can be easily identified in advance. Notably, this condition inevitably leads to a singular configuration—see Section 4.2, which can thus be safely disregarded. Nevertheless, up to four MP configurations can still be determined, with the calculation procedure provided in the supplementary material for interested readers.

- (3) $\{\cot \vartheta_{vi}\}_1^3$ or $\{\cot \vartheta_{wi}\}_1^3$ will approach infinity when the i th intermediate- or distal-joint axis becomes parallel to S —the intersection line of planes Π_v and Π_w . Notably, ϑ_{v1} can still be solved using Eqs. (19) or (25) even if any of $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{w}_2$ or $\mathbf{w}_3$ is parallel to S , after which the remaining unknowns can be computed as usual. However, if $\mathbf{v}_1$ or $\mathbf{w}_1$ is parallel to S , E will inevitably vanish, a case to be discussed in the next bullet. The detailed proof is omitted due to space limitations but is provided in the supplementary material.
- (4) As for the vanishing of denominators, this may occur in Eqs. (19) or (25), which happens iff $E = 0$. Notably, this case can be divided into two subcases depending on the value of F :
- $E = 0$ and $F = 0$, which occurs iff $\cot v_{12}/\cot \omega_{12} = \cot v_{13}/\cot \omega_{13} = \pm 1$, a case that can be easily identified in advance. Notably, this inevitably leads to singularities—see Section 4.2, which can thus be safely disregarded. However, up to two MP configurations can still be determined in this case, whose calculation procedure is provided in the supplementary material.
 - $E = 0$ and $F \neq 0$, which can be easily handled by converting Eqs. (19) or (25) into the form $\tan(2\vartheta_{v1}) = -2\zeta_2/\zeta_1 = E/F = 0$, yielding four possible solutions for ϑ_{v1} , namely, $\vartheta_{v1} = k\pi/2$, with $k \in \{0, 1, 2, 3\}$. The remaining unknowns can then be calculated as usual.

Lastly, it should be noted that according to Eq. (9), $\cos \phi$ can be computed using three distinct expressions, namely,

$$\cos \phi = -\cot \vartheta_{w1} \cot \vartheta_{v1} = -\cot \vartheta_{w2} \cot \vartheta_{v2} = -\cot \vartheta_{w3} \cot \vartheta_{v3} \quad (30)$$

Moreover, the latter two expressions become relevant when the first expression fails to yield the correct value of $\cos \phi$, which occurs when $\vartheta_{v1} = k\pi$, with $k \in \mathbb{Z}$. Additionally, although the aforementioned formulations lead to a linear univariate polynomial in $\cot(2\vartheta_{v1})$, similar univariate polynomials in $\cot(2\vartheta_{v2})$ or $\cot(2\vartheta_{v3})$ can be obtained by simply exchanging subscripts 1, 2, 3 with 2, 3, 1 or 3, 1, 2 in the proposed formulations. These alternative univariate polynomials can be used when Eqs. (19) or (25) fail to produce the correct solutions for $\cot(2\vartheta_{v1})$.

3.4. Case study

Next, a representative case study is conducted to validate the proposed FK formulations. Specifically, the architectural parameters of the MP are arbitrarily selected as $[\gamma_1, \gamma_2, \gamma_3] = [5\pi/9, 2\pi/3, 7\pi/9]$, while the input joint variables are specified as $\boldsymbol{\theta} = [-0.95, -3.26, -5.13]^T$.³ Then, the corresponding linear univariate polynomial, as given by Eqs. (19) or (25), is computed as

$$\cot(2\vartheta_{v1}) = 1.3612202 \quad (31)$$

from which four solutions for ϑ_{v1} are obtained, namely,

$$\vartheta_{v1} = 0.3168, \quad 1.8876, \quad 3.4584, \quad 5.0292 \quad (32)$$

Subsequently, the remaining unknowns can be calculated following the procedure described in Section 3.2, which are all listed in Table 1.

As expected, only four sets of these FK solutions are valid, as identified using the condition $|\cos \phi| \leq 1$. Moreover, these four valid sets share the same value of ϕ , with ψ and σ either identical or differing by π . Their corresponding robot configurations are shown in Fig. 3, where the black arrow represents the normal vector ($\mathbf{n}_w$) of the MP, and the black dashed line denotes the line of intersection between planes Π_v and Π_w , labeled as S . As observed, the angle between $\mathbf{n}_w$ and the z -axis remains constant across these configurations, and all configurations share the same S . These observations are consistent with the statement made at the beginning of Section 3.

Finally, it is worth noting that the correctness of the FK solutions has been verified using the formulation proposed by Gosselin et al. [6] and that developed in our earlier work [15], along with the inverse kinematics of the robot [1]. Additionally, we have conducted a large number of similar numerical tests, which thoroughly validate the proposed FK formulations. Given the straightforward nature of these verifications, the details are omitted due to space limitations.

4. Singularity analysis

In this section, the singularity analysis of the robot class is performed in both the MP orientation workspace and the input joint space. Notably, the robot class exhibits minimal singularity loci that are both intuitive and physically interpretable, greatly facilitating tasks such as singularity avoidance and path planning. Furthermore, the analysis reveals the potential for a large singularity-free orientation workspace, thereby enhancing the overall performance of the robot class.

To facilitate the subsequent singularity analysis, we first introduce the velocity-level kinematics relations of the robot class under study [17], namely,

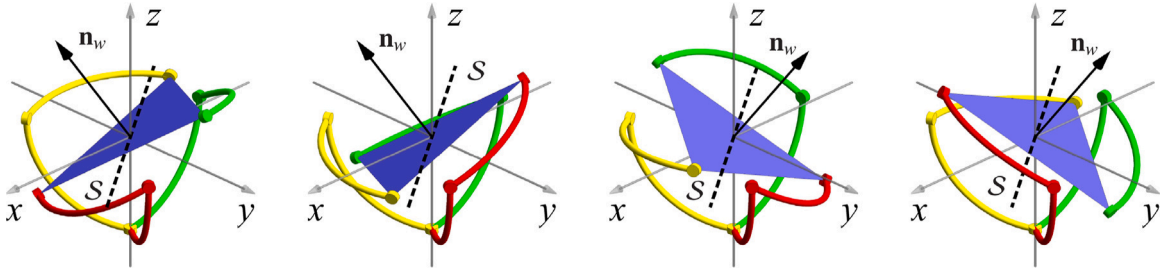
$$\mathbf{H}\boldsymbol{\omega} = \mathbf{D}\dot{\boldsymbol{\theta}} \quad (33)$$

³ Unless otherwise specified, all angles presented in this paper are expressed in radians.

Table 1

FK solutions for the case study.

Sol	θ_{v1}	θ_{v2}	θ_{v3}	θ_{w1}	θ_{w2}	θ_{w3}	$\cos \phi$	ϕ	ψ	σ
1	0.3168	2.6268	4.4968	4.9755	1.1358	2.8810	0.8216	0.6067	5.3456	5.6087
2	0.3168	2.6268	4.4968	1.8339	4.2774	6.0227	0.8216	0.6067	5.3456	2.4671
3	3.4584	5.7684	1.3552	4.9755	1.1358	2.8810	0.8216	0.6067	2.2040	2.4671
4	3.4584	5.7684	1.3552	1.8339	4.2774	6.0227	0.8216	0.6067	2.2040	5.6087
5	1.8876	4.1976	6.0676	0.2631	2.7066	4.4519	1.2172	–	–	–
6	1.8876	4.1976	6.0676	3.4047	5.8481	1.3103	1.2172	–	–	–
7	5.0292	1.0560	2.9260	0.2631	2.7066	4.4519	1.2172	–	–	–
8	5.0292	1.0560	2.9260	3.4047	5.8481	1.3103	1.2172	–	–	–

**Fig. 3.** The robot configurations corresponding to the four sets of valid FK solutions in Table 1.

where $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T$ denotes the angular velocity of the MP, and $\dot{\boldsymbol{\theta}} = [\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3]$ denotes the array of input joint variables. Moreover, $\mathbf{H}$ and $\mathbf{D}$, named as the forward and inverse Jacobian matrices, respectively, can be derived as [17]

$$\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3]^T, \quad \mathbf{D} = \text{diag}[d_1, d_2, d_3], \quad \text{with } \mathbf{h}_i = \mathbf{v}_i \times \mathbf{w}_i \text{ and } d_i = \mathbf{u}_i \cdot \mathbf{h}_i \quad (34)$$

for $i = 1, 2, 3$. Then, the inverse and forward singularities arise when the determinants of $\mathbf{D}$ and $\mathbf{H}$ vanish, respectively, which can be characterized by [17]

$$d_1 \cdot d_2 \cdot d_3 = 0 \text{ and } (\mathbf{h}_1 \times \mathbf{h}_2) \cdot \mathbf{h}_3 = 0 \quad (35)$$

respectively. Based on these conditions, the inverse and forward singularities are subsequently analyzed.

4.1. Singularity loci in the MP orientation workspace

In order to obtain the singularity loci in the MP orientation workspace, we first express $\{\mathbf{u}_i\}_1^3$ and $\{\mathbf{w}_i\}_1^3$, as appearing in $\mathbf{H}$ and $\mathbf{D}$ (Eq. (34)), in terms of the azimuth, tilt, and torsion angles of the MP orientation. Specifically, $\{\mathbf{u}_i\}_1^3$ — the unit direction vectors associated with the three coaxial input-joint axes — can be simply determined as

$$[\mathbf{u}_i]_B = [0, 0, -1]^T, \quad \text{for } i = 1, 2, 3 \quad (36)$$

where all these vectors are expressed in the base frame $\mathcal{F}_B$. As for $\{\mathbf{w}_i\}_1^3$ — the unit direction vectors associated with the three distal-joint axes — can be readily computed from Eqs. (1) and (2), namely,

$$[\mathbf{w}_i]_B = \mathbf{Q}(\psi, \phi, \sigma)[\mathbf{w}_{i0}]_B \\ = [c(\psi)c(\phi)c(\delta_i) - s(\psi)s(\sigma - \psi), c(\psi)s(\delta_i) + s(\psi)c(\phi)c(\delta_i), -s(\phi)c(\delta_i)]^T, \quad \text{for } i = 1, 2, 3 \quad (37)$$

where $\delta_1 \equiv \sigma - \psi$, $\delta_2 \equiv \sigma - \psi + \gamma_3$, and $\delta_3 \equiv \sigma - \psi - \gamma_2$.

4.1.1. Inverse singularity

As derived in Eq. (35), the inverse singularity occurs iff at least one of $\{d_i\}_1^3$ vanishes. This leads to

$$d_i = \mathbf{u}_i \cdot \mathbf{h}_i = \mathbf{u}_i \cdot (\mathbf{v}_i \times \mathbf{w}_i) = 0, \quad \text{for some } i, \quad i \in \{1, 2, 3\} \quad (38)$$

implying that $\mathbf{u}_i$, $\mathbf{v}_i$ and $\mathbf{w}_i$ become coplanar for at least one of the limbs. After further considering that $\mathbf{v}_i \perp \mathbf{u}_i$ and $\mathbf{v}_i \perp \mathbf{w}_i$, it can be readily derived from Eq. (38) that $\mathbf{u}_i \parallel \mathbf{w}_i$ or, equivalently,

$$\mathbf{u}_i \cdot \mathbf{w}_i = \pm 1, \quad \text{for some } i, \quad i \in \{1, 2, 3\} \quad (39)$$

After substituting Eqs. (36) and (37) into Eq. (39), we obtain

$$\begin{cases} -\sin \phi \cos(\sigma - \psi) = \pm 1, \text{ or} \\ -\sin \phi \cos(\sigma - \psi + \gamma_3) = \pm 1, \text{ or} \\ -\sin \phi \cos(\sigma - \psi - \gamma_2) = \pm 1 \end{cases} \quad (40)$$

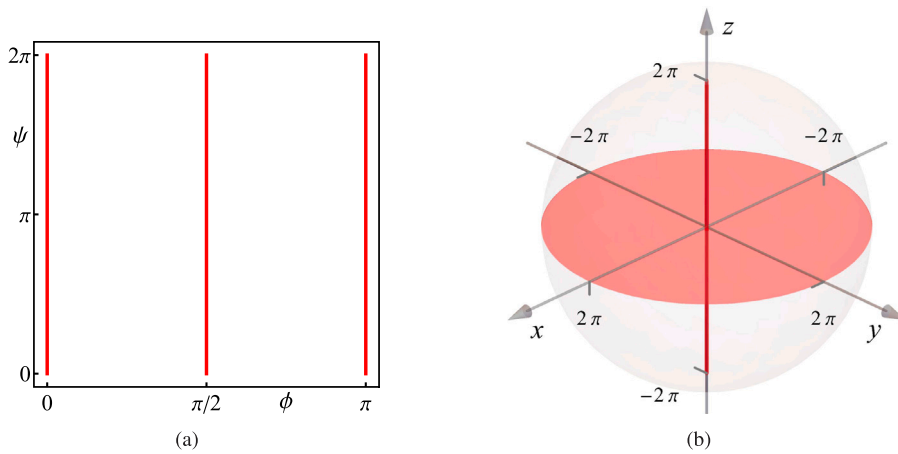


Fig. 4. The singularity loci in the MP orientation workspace, applicable to any instance within the robot class, plotted in (a) the two-dimensional tilt and azimuth space, and (b) a three-dimensional spherical coordinate system, where $[x, y, z] \equiv \sigma[\cos \psi \sin \phi, \sin \psi \sin \phi, \cos \phi]$.

or, equivalently,

$$\begin{cases} \phi = \pi/2 & \text{and } \sigma - \psi = k\pi, \text{ or} \\ \phi = \pi/2 & \text{and } \sigma - \psi = -\gamma_3 + k\pi, \text{ or} \\ \phi = \pi/2 & \text{and } \sigma - \psi = \gamma_2 + k\pi \end{cases} \quad \text{with } k \in \mathbb{Z} \quad (41)$$

which fully characterizes the inverse singularity of the robot class in the MP orientation workspace. Geometrically, these conditions imply that the inverse singularity occurs iff at least one of the limbs becomes either fully folded or fully extended. Alternatively, from the perspective of MP orientation, the inverse singularity arises iff $\mathbf{n}_w$ lies within the x - O - y plane, and the azimuth and torsion angles of the MP satisfy the specific relationships given in Eq. (41).

4.1.2. Forward singularity

As indicated by Eq. (35), the forward singularity occurs iff

$$(\mathbf{h}_1 \times \mathbf{h}_2) \cdot \mathbf{h}_3 = 0 \quad (42)$$

where

$$\mathbf{h}_i = \mathbf{v}_i \times \mathbf{w}_i \parallel (\mathbf{w}_i \times \mathbf{u}_i) \times \mathbf{w}_i = (\mathbf{w}_i \cdot \mathbf{w}_i)\mathbf{u}_i - \mathbf{w}_i(\mathbf{u}_i \cdot \mathbf{w}_i) = \mathbf{u}_i - (\mathbf{u}_i \cdot \mathbf{w}_i)\mathbf{w}_i \quad (43)$$

Then, the singularity condition can be readily derived upon substituting Eqs. (43), (36) and (37) into Eq. (42), namely,

$$[(\mathbf{u}_1 - (\mathbf{u}_1 \cdot \mathbf{w}_1)\mathbf{w}_1) \times (\mathbf{u}_2 - (\mathbf{u}_2 \cdot \mathbf{w}_2)\mathbf{w}_2)] \cdot (\mathbf{u}_3 - (\mathbf{u}_3 \cdot \mathbf{w}_3)\mathbf{w}_3) = -\cos \phi \sin^2 \phi \sin \gamma_1 \sin \gamma_2 \sin \gamma_3 = 0 \quad (44)$$

where none of $\{\sin \gamma_i\}_1^3$ vanishes—see footnote (1). This means that, for the robot class under study, the forward singularity occurs iff

$$\phi = k\pi/2, \quad \text{with } k = 0, 1, 2 \quad (45)$$

regardless of the angles between the three coplanar distal-joint axes on the MP.

Interestingly, it can be readily seen that the conditions for the inverse singularity, as derived in Eq. (41), are fully encompassed by those for the forward singularity. This implies that the conditions displayed in Eq. (45) are sufficient to characterize both the forward and inverse singularities for any instance of the robot class, regardless of the layout of the three axes on the MP. Now, the singularity loci in the MP orientation workspace, as characterized by Eq. (45), are plotted in Fig. 4 in two different representations, the first in the two-dimensional tilt and azimuth space (since the torsion is irrelevant here), and the second in a three-dimensional spherical coordinate system, where $[x, y, z] \equiv \sigma[\cos \psi \sin \phi, \sin \psi \sin \phi, \cos \phi]$. It is noteworthy that the second representation is introduced to aid in the visualization of the direction of the normal vector of the MP and the corresponding torsion.

As can be seen from Fig. 4, the robot class encounters singularities iff $\mathbf{n}_w$ (the normal vector of the MP) either aligns with the z -axis or lies within the x - O - y plane. Moreover, these figures also suggest that the robot class can theoretically perform infinite torsional motion, as long as $\mathbf{n}_w$ is not oriented vertically or horizontally, revealing the potential for the robot class to achieve a large singularity-free orientation workspace. This characteristic makes the robot class particularly promising for applications requiring infinite torsional motion, such as orienting devices, robotic wrists and shoulders [1,2].

4.2. Singularity loci in the input joint space

As demonstrated in Section 4.1, the robot class encounters singularities iff $\phi = k\pi/2$, with $k = 0, 1, 2$. Building upon these conditions, this section derives the corresponding singularity conditions in the input joint space, which enable the early detection of singularities when performing the FK computations.

First, we deal with the case $\phi = 0$. Upon substituting this condition into Eq. (37), the triad $\{\mathbf{w}_i\}_1^3$, when expressed in $\mathcal{F}_B$, simplifies to

$$[\mathbf{w}_i]_B = [\cos \epsilon_i, \sin \epsilon_i, 0]^T, \quad \text{for } i = 1, 2, 3 \quad (46)$$

where $\epsilon_1 \equiv \sigma$, $\epsilon_2 \equiv \sigma + \gamma_3$, and $\epsilon_3 \equiv \sigma - \gamma_2$. Accordingly, the triad $\{\mathbf{v}_i\}_1^3$ associated with the three intermediate-joint axes can be obtained as

$$[\mathbf{v}_i]_B = \pm[\mathbf{u}_i]_B \times [\mathbf{w}_i]_B = \pm[\sin \epsilon_i, -\cos \epsilon_i, 0]^T, \quad \text{for } i = 1, 2, 3 \quad (47)$$

Moreover, $\{\mathbf{v}_i\}_1^3$ can also be expressed in terms of the input joint variables, namely,

$$[\mathbf{v}_i]_B = [\cos \theta_i, -\sin \theta_i, 0]^T, \quad \text{for } i = 1, 2, 3 \quad (48)$$

After combining Eqs. (47) and (48), θ_i can be readily determined as

$$\theta_i = (2k_i + 1)\pi/2 - \epsilon_i, \quad \text{for } i = 1, 2, 3, \quad \text{with } k_i \in \mathbb{Z} \quad (49)$$

where ϵ_i can take arbitrary values. Therefore, the condition given by Eq. (49) is equivalent to the following relationships among the input joint variables:

$$\begin{aligned} v_{12} &\equiv \theta_1 - \theta_2 = (k_1 - k_2)\pi + \gamma_3 = (k_1 - k_2)\pi + \omega_{12} \\ v_{13} &\equiv \theta_1 - \theta_3 = (k_1 - k_3)\pi - \gamma_2 = (k_1 - k_3)\pi + \omega_{13} \end{aligned} \quad (50)$$

or, equivalently,

$$\cot v_{12} / \cot \omega_{12} = \cot v_{13} / \cot \omega_{13} = 1 \quad (51)$$

This equation represents the singularity condition in the input joint space corresponding to the case $\phi = 0$.

Similarly, for the case $\phi = \pi$, the corresponding singularity condition can be readily derived in the input joint space using a similar procedure, leading to

$$\begin{aligned} v_{12} &= \theta_1 - \theta_2 = (k_1 - k_2)\pi - \gamma_3 = (k_1 - k_2)\pi - \omega_{12} \\ v_{13} &= \theta_1 - \theta_3 = (k_1 - k_3)\pi + \gamma_2 = (k_1 - k_3)\pi - \omega_{13} \end{aligned} \quad (52)$$

or, equivalently,

$$\cot v_{12} / \cot \omega_{12} = \cot v_{13} / \cot \omega_{13} = -1 \quad (53)$$

Finally, we consider the case $\phi = \pi/2$, under which the normal vector ($\mathbf{n}_w$) of the MP lies within the x - O - y plane. After some straightforward geometric reasoning, it can be readily seen that this case occurs iff at least two of $\{\mathbf{v}_i\}_1^3$ become collinear or, equivalently,

$$v_{ij} = \theta_i - \theta_j = k\pi \quad \text{for some } i, j \in \{1, 2, 3\}, \quad \text{with } i \neq j \text{ and } k \in \mathbb{Z} \quad (54)$$

To this end, Eqs. (51), (53) and (54) establish the singularity conditions in the input joint space, which, when interpreted geometrically, imply that the robot class encounters singularities iff (a) the angles between each pair of the three intermediate-joint axes are identical to those between the corresponding pair of the distal-joint axes, or (b) at least two of the intermediate-joint axes become collinear. These concise conditions enable the early detection of singularities when performing FK computations, as discussed in Section 3.3, thereby facilitating the simulation and feedback control of the robot class.

5. Dexterity analysis

In this section, the dexterity of the 3-RRR robot class under study is investigated. First, the distributions of the local dexterity index (LDI) for three representative instances are examined, revealing that the LDI distribution varies with the architectural parameters, even though the robot class exhibits identical singularity loci. Based on this observation, the influence of these parameters on the global dexterity index (GDI) is further analyzed, providing guidelines for selecting architectural parameters to enhance the robot's operational performance from the dexterity perspective.

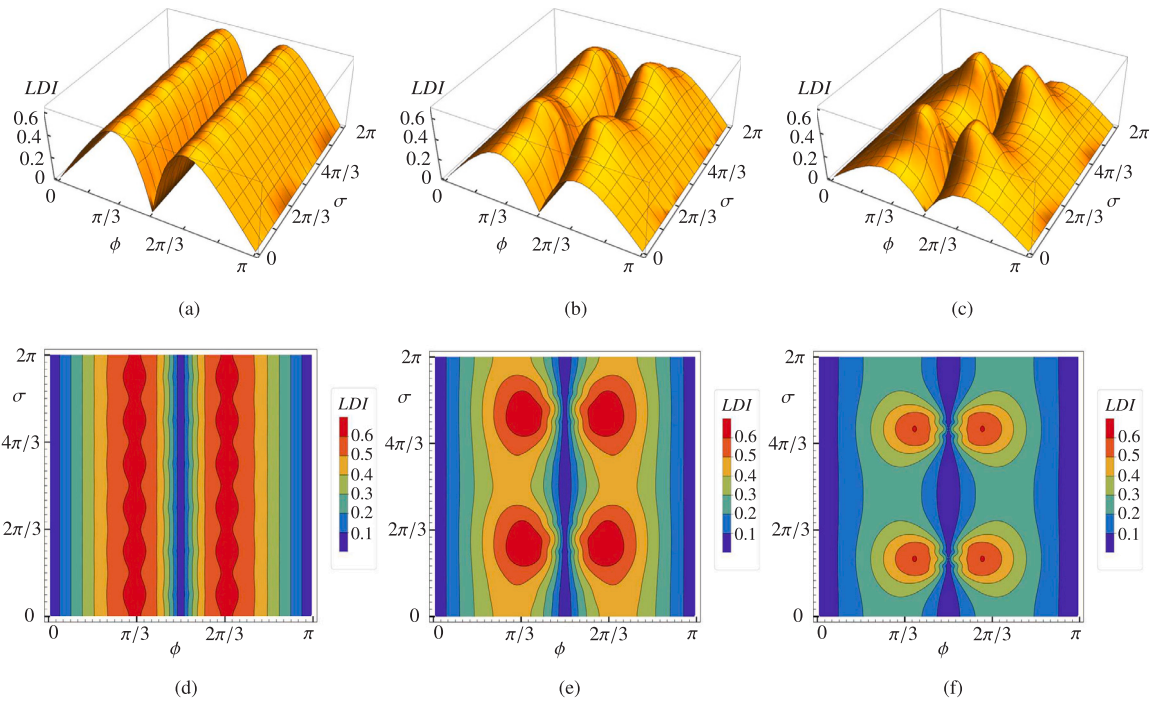


Fig. 5. The LDI over $\phi \in [0, \pi]$ and $\sigma \in [0, 2\pi]$ for three representative 3-RRR SPMs from the robot class, characterized by (a) $[\gamma_1, \gamma_2, \gamma_3] = [2\pi/3, 2\pi/3, 2\pi/3]$, (b) $[\gamma_1, \gamma_2, \gamma_3] = [5\pi/9, 2\pi/3, 7\pi/9]$, and (c) $[\gamma_1, \gamma_2, \gamma_3] = [4\pi/9, 4\pi/9, 10\pi/9]$. Subfigures (d)–(f) show their corresponding contoured representations.

5.1. The distribution of the LDI within the orientation workspace

To evaluate the local dexterity of the robot class under study, we adopt a commonly used local performance metric — the reciprocal of the condition number of the Jacobian matrix — as the LDI [31–33], namely,

$$\text{LDI} = 1/\kappa(\mathbf{J}) \quad (55)$$

where the condition number $\kappa(\mathbf{J})$ is computed based the Frobenius-norm of the Jacobian matrix [31–33], namely,

$$\kappa(\mathbf{J}) = \frac{1}{n} \sqrt{\text{tr}(\mathbf{J}\mathbf{J}^T)\text{tr}[(\mathbf{J}\mathbf{J}^T)^{-1}]} = \frac{1}{n} \sqrt{\|\mathbf{J}\|_F^2 \|\mathbf{J}^{-1}\|_F^2}, \quad n = 3 \quad (56)$$

with $\mathbf{J} = \mathbf{D}^{-1}\mathbf{H}$. The LDI, as defined above, ranges from 0 to 1, with higher values indicating better dexterity.

Interestingly, owing to the rotational symmetry of the robot class about the z -axis,⁴ the LDI distribution, though inherently defined in the three-dimensional orientation workspace, can be confined to a two-dimensional subspace defined by the tilt and torsion angles, while the azimuth angle can be arbitrarily specified. This simplification helps reduce computational cost and has been widely adopted in the analysis of SPMs with coaxial input-joint axes [17,43]. In this vein, the azimuth angle is set to $\pi/2$ in the following evaluation, aligning the MP tilt axis with the x -axis.

Fig. 5 illustrates the LDI distribution over the subspace $\{\phi, \sigma\}$ for three representative 3-RRR SPMs from the robot class, each defined by a distinct set of $\{\gamma_i\}_1^3$ —the angles between the three coplanar distal-joint axes on the MP. These figures demonstrate that the LDI distribution is symmetric about the plane $\phi = \pi/2$ for all instances of the robot class. Moreover, the LDI always vanishes at $\phi = 0, \pi/2$, and π , consistent with the results of the singularity analysis. Furthermore, Fig. 5 indicates that although the robot class exhibits identical singularity loci, the LDI distribution over the workspace varies with the architectural parameters of the MP.

⁴ By this we mean that, owing to the alignment of the three input-joint axes with the z -axis, the relative orientations of the robot's nine axes remain unchanged when the entire robot rotates about the z -axis, thereby keeping the LDI invariant.

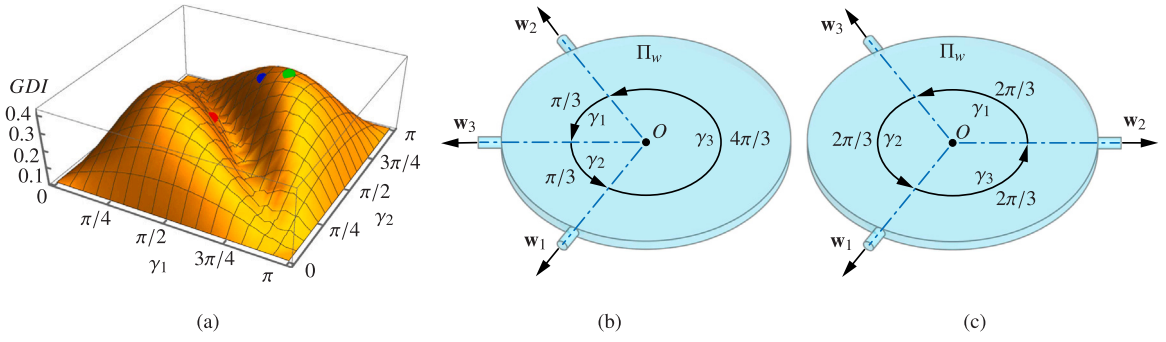


Fig. 6. (a) The GDI over $\gamma_1 \in (0, \pi)$ and $\gamma_2 \in (0, \pi)$ for the robot class under study, and the two MP architectures achieving optimal GDI, characterized by (b) $[\gamma_1, \gamma_2, \gamma_3] = [\pi/3, \pi/3, 4\pi/3]$ and (c) $[\gamma_1, \gamma_2, \gamma_3] = [2\pi/3, 2\pi/3, 2\pi/3]$, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.2. The influence of the architectural parameters on the GDI

To further investigate the influence of the architectural parameters on the robot dexterity, the GDI is introduced to quantify the overall dexterity performance of the robot, defined as follows [33]:

$$\text{GDI} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\kappa_i(\mathbf{J})} \quad (57)$$

where N denotes the number of discrete points sampled from the workspace, and $\kappa_i(\mathbf{J})$ is the condition number of the Jacobian matrix corresponding to the i th discrete point. Moreover, the computational efficiency of the evaluation process can be improved by exploiting the rotational symmetry of the robot class about the z -axis, as well as the symmetry of the LDI distribution about the plane $\phi = \pi/2$, as discussed in Section 5.1. Therefore, the GDI evaluation is confined to the ranges $\phi \in [0, \pi/2]$ and $\sigma \in [0, 2\pi)$, with both variables discretized with an interval of $\pi/36$, thus yielding $N = 19 \times 72 = 1368$ discrete points characterizing the workspace. Then, the GDI for a given instance of the robot class can be computed using Eq. (57).

Next, the influence of the MP architectural parameters $\{\gamma_i\}_1^3$ on the GDI is examined. Given that $\gamma_1 + \gamma_2 + \gamma_3 = 2\pi$ and only two of these parameters are independent, the GDI is evaluated over the ranges $\gamma_1 \in (0, \pi)$ and $\gamma_2 \in (0, \pi)$, as illustrated in Fig. 6(a). Interestingly, it can be observed that the GDI is symmetrically distributed about the plane $\gamma_1 + \gamma_2 = \pi$ and the plane $\gamma_1 - \gamma_2 = 0$. Moreover, the GDI approaches zero when γ_i approaches $k\pi$, for some $i \in \{1, 2, 3\}$ and $k \in \mathbb{Z}$. This implies that the MP architectures with two of $\{\mathbf{w}_i\}_1^3$ aligned will inevitably lead to singularities, aligning with the findings obtained in Section 4.1.2. Furthermore, the green, blue and red points in Fig. 6(a) correspond to the three design examples in Fig. 5, with GDI values of approximately 0.425, 0.364 and 0.239, respectively, which are broadly consistent with their LDI performance illustrated in Fig. 5. Finally, Fig. 6(a) also indicates that the robot class under study achieves optimal GDI when $\gamma_1 = \gamma_2 = \pi/3$ or $\gamma_1 = \gamma_2 = 2\pi/3$, both reaching approximately 0.425, with the corresponding MP layout shown in Figs. 6(b) and 6(c).

It is worth noting that the kinematic performance of the robot class has also been evaluated using the local transmission index (LTI) and global transmission index (GTI) — two commonly used screw-theory-based metrics with clear physical interpretations [29, 56] — following a procedure similar to that based on the LDI and GDI. The results show that the distributions of the LTI and GTI are generally similar to those of the LDI and GDI in terms of both magnitude and trend. Notably, the GTI and GDI attain their optimal values at the same MP architectural parameters, namely, $\gamma_1 = \gamma_2 = \pi/3$ or $\gamma_1 = \gamma_2 = 2\pi/3$. Due to space limitations, the details of these analyses are omitted from the manuscript but are provided in the supplementary material for interested readers.

Lastly, it is also worth noting that the proposed FK formulations, together with the singularity and dexterity distributions, can also be applied to a variety of SPMs, provided that they are kinematically equivalent to 3-RRR SPMs [14,15,17,42,45] and that their geometries are tuned such that their 3-RRR kinematic equivalents fall within the robot class under study.

6. Workspace analysis

In this section, the workspace analysis is conducted for the proposed 3-RRR robot class. Interestingly, all instances of this class will be shown to share the same singularity-free orientation workspace when mechanical interference is not considered, allowing the MP to undergo infinite torsional motion around its normal vector ($\mathbf{n}_w$), provided that this vector is not oriented vertically or horizontally w.r.t the base frame. Moreover, it will be demonstrated that the effect of mechanical interference can be almost completely eliminated when the link shapes are appropriately specified. As a result, the robot class can achieve a significantly large tilt-torsion orientation workspace, enabling infinite torsional motion when pointed within nearly a hemisphere, making it a promising candidate for various applications.

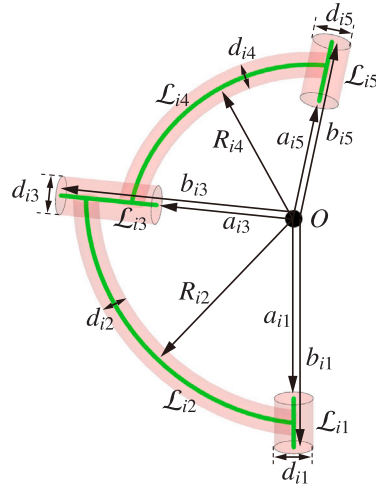


Fig. 7. The design of the i th limb and the associated notations.

6.1. The evaluation of the orientation workspace

First, we prove that all instances of the 3-RRR robot class share an identical singularity-free orientation workspace when mechanical interference is neglected, allowing infinite torsional motion when pointed within nearly a hemisphere. As is well known, the workspace of a PM can be obtained as the intersection of the workspaces of its individual limbs when singularities and mechanical interference are omitted. Moreover, given that the robot class consists solely of revolute joints, we can reasonably assume that the motion ranges of these joints are unconstrained. After further considering the axis arrangements of the robot class introduced in Section 2, it can be readily derived that the orientation workspaces of all its limbs — and thus their intersection — fully span the entire three-dimensional orientation space. Furthermore, since the robot class encounters singularities iff the normal vector of the MP either aligns with the z -axis or lies within the x - O - y plane, as demonstrated in Fig. 4, it can be readily concluded that all instances of the robot class admit the same singularity-free orientation workspace, allowing infinite torsional motion when pointed within nearly a hemisphere, provided that mechanical interference is neglected.

Next, we consider the issue of mechanical interference. In principle, such interference may occur among the BP, the MP, and any components within the three limbs. However, by following simple design strategies available in the literature [21,23,45–47], interference involving the BP and MP, as well as interference between components within the same limb, can be inherently eliminated. More specifically, since all components of the limbs and the MP undergo pure spherical motion with respect to the BP about the spherical center, each is confined to a specific spherical layer centered at this point. By designing the BP and MP so that their bodies lie entirely outside or inside the spherical layers occupied by the limbs, interference involving the BP and MP can be avoided. Similarly, interference between components within the same limb can be eliminated by placing these components in distinct spherical layers. To this end, only limb-to-limb interference remains relevant. Apparently, such mechanical interference depends not only on the architectural parameters of the robot but also on the specific shapes of the links, the latter admitting diverse designs and thus not uniquely defined.

For the sake of simplicity, we assume that each limb of the robot class comprises five components: two arc-shaped tubes with uniform circular cross-sections representing the proximal and distal links, and three cylinders enveloping the proximal, intermediate, and distal joints, as illustrated in Fig. 7. As will be shown later, this design is sufficient to eliminate the effects of mechanical interference of the robot class when the link shapes and architectural parameters are appropriately specified. Next, the geometries of these components are specified as follows: For the j th component of the i th limb, its central axis and cross-sectional diameter are denoted as $\mathcal{L}_{ij}$ and d_{ij} , respectively, for $i = 1, 2, 3$ and $j = 1, 2, \dots, 5$. Here, $\mathcal{L}_{ij}$ corresponds to a circular arc in the case of tubes and a line segment in the case of cylinders. Moreover, the radii of circular arcs $\mathcal{L}_{i2}$ and $\mathcal{L}_{i4}$ are denoted as R_{i2} and R_{i4} , respectively. Furthermore, for the line segment $\mathcal{L}_{ij}$, the distances from its two endpoints to the rotational center O are denoted as a_{ij} and b_{ij} , respectively, for $i = 1, 2, 3$ and $j = 1, 3, 5$.

Now, we are ready to establish the constraints induced by mechanical interference. Under the aforementioned designs, mechanical interference of the studied 3-RRR robot class can be categorized into intra-limb and inter-limb interference; moreover, the former can be eliminated by simply placing the arc-shaped proximal and distal links of the same limb into two spherical layers of different radii [23], which can be characterized by the condition:

$$|R_{i2} - R_{i4}| \geq (d_{i2} + d_{i4})/2, \quad \text{for } i = 1, 2, 3 \quad (58)$$

After that, only inter-limb interference needs to be considered, leading to the following condition:

$$\text{distance}(\mathcal{L}_{ij}, \mathcal{L}_{mn}) \geq (d_{ij} + d_{mn})/2, \quad \text{for } i, m = 1, 2, 3, \quad j, n = 1, 2, \dots, 5, \quad \text{and } i \neq m \quad (59)$$

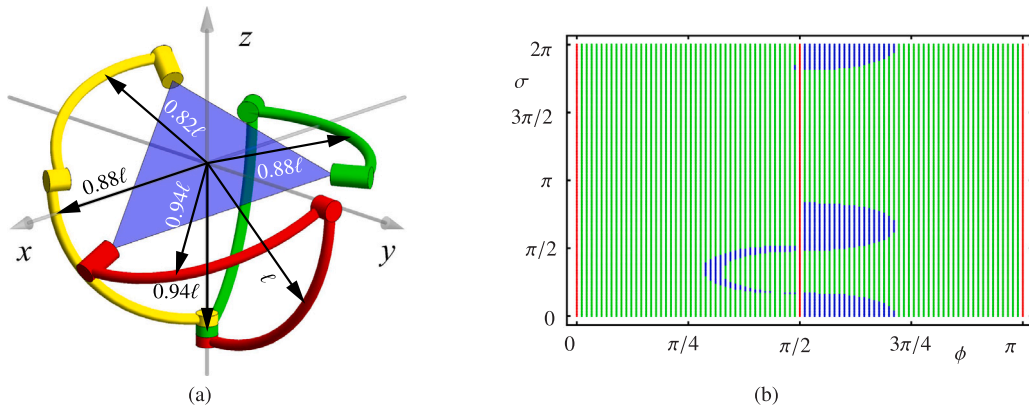


Fig. 8. (a) A specific design of the robot class under study, incorporating a symmetric MP and six arc-shaped links with radii of $[R_{12}, R_{22}, R_{32}, R_{14}, R_{24}, R_{34}] = [\ell, 0.94\ell, 0.88\ell, 0.94\ell, 0.88\ell, 0.82\ell]$, and (b) its corresponding orientation workspace in the subspace $\{\phi, \sigma\}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where $\text{distance}(\mathcal{L}_{ij}, \mathcal{L}_{mn})$ represents the minimum distance between $\mathcal{L}_{ij}$ and $\mathcal{L}_{mn}$, which can be found using the approach provided in the supplementary material.

Next, we introduce a discretization method for the workspace evaluation, incorporating the constraints induced by singularities and mechanical interference, as described earlier in this section. First, similar to the evaluation of the LDI distribution, the rotational symmetry of the robot class about the z -axis (as explained in footnote 5) allows the workspace evaluation to be confined to a two-dimensional subspace defined by $\phi \in [0, \pi]$ and $\sigma \in [0, 2\pi]$, with the azimuth angle ψ arbitrarily specified as $\pi/2$. Next, this subspace is discretized with an interval of $\pi/90$ ⁵ along both dimensions, yielding a total of $91 \times 181 = 16471$ discrete points. Among these points, those with $\phi = 0, \pi/2$, or π are discarded, as they correspond to singular configurations identified in Section 4.1. For the remaining points, the inverse kinematics [1] is performed to obtain the corresponding robot configurations,⁶ which are then examined for potential mechanical interference using Eq. (59). After further discarding the points that violate the interference constraints, the orientation workspace of the robot can be finally obtained.

6.2. Case study

Next, a case study is conducted to examine the orientation workspace of the robot class under study. Specifically, the MP architecture shown in Fig. 6(c) is adopted here, as it not only achieves optimal dexterity performance but also exhibits a symmetric architecture, both of which are generally preferred in the design of PMs [57,58]. Moreover, the parameters related to the link shapes introduced in Section 6.1 are respectively specified as

$$\begin{aligned} [d_{i1}, d_{i2}, d_{i3}, d_{i4}, d_{i5}] &= [0.12\ell, 0.06\ell, 0.12\ell, 0.06\ell, 0.12\ell] \\ [R_{12}, R_{22}, R_{32}, R_{14}, R_{24}, R_{34}] &= [\ell, 0.94\ell, 0.88\ell, 0.94\ell, 0.88\ell, 0.82\ell] \\ [a_{i1}, a_{i3}, a_{i5}, b_{i1}, b_{i3}, b_{i5}] &= [R_{i2} - d_{i2}/2, R_{i4} - d_{i4}/2, R_{i4} - 3d_{i4}, R_{i2} + d_{i2}/2, R_{i2} + d_{i2}/2, R_{i4} + d_{i4}/2] \end{aligned} \quad (60)$$

where ℓ denotes the maximum radius among the arc-shaped central axes of the six links, serving as a representative measure of the robot size. The above parameters guarantee that the three arc-shaped proximal links of different limbs are assigned into distinct layers, and that the proximal and distal links of each limb are separated into different layers, ultimately resulting in the specific design of the 3-RRR SPM shown in Fig. 8(a). The corresponding workspace of the robot in the subspace $\{\phi, \sigma\}$ is plotted in Fig. 8(b), where the green points represent the feasible configurations, while the red and blue points represent the invalid configurations induced by singularities and mechanical interference, respectively. As can be seen, mechanical interference significantly restricts the workspace of the robot when the tilt angle ϕ is around 90° . As a result, the robot can only generate infinite torsional motion when $\phi \in [2^\circ, 50^\circ]$ or $\phi \in [130^\circ, 178^\circ]$, significantly narrower than the theoretical range of ϕ illustrated in Fig. 4.

After further examination, it is found that the aforementioned mechanical interference arises from the arc-shaped links of different limbs being located in the same spherical layers, specifically characterized by $R_{14} = R_{22}$ and $R_{24} = R_{32}$. To further mitigate this interference, we rearrange the six arc-shaped links into six different layers according to the following geometric parameters:

$$[R_{12}, R_{22}, R_{32}, R_{14}, R_{24}, R_{34}] = [\ell, 0.94\ell, 0.88\ell, 0.82\ell, 0.76\ell, 0.70\ell] \quad (61)$$

⁵ Notably, three discretization resolutions — $\pi/45$, $\pi/90$, and $\pi/180$ — were tested initially, showing limited differences in the resulting workspace. Therefore, we selected $\pi/90$ as a reasonable trade-off between computational cost and representational accuracy, which is consistent with values commonly adopted in the literature [17,23].

⁶ Notice that the inverse kinematics of a 3-RRR SPM admits at most eight solutions, corresponding to eight distinct working modes of the robot. Here, we assume that the robot class under study operates within the working mode illustrated in Fig. 1, characterized by the conditions that $(\mathbf{u}_i \times \mathbf{v}_i) \cdot \mathbf{w}_i > 0$, for $i = 1, 2, 3$.

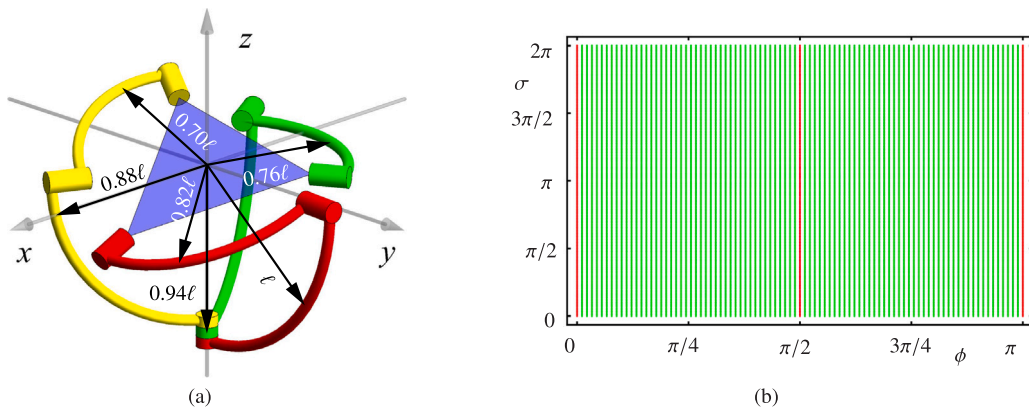


Fig. 9. (a) A specific design of the robot class under study, incorporating a symmetric MP and six arc-shaped links with radii of $[R_{12}, R_{22}, R_{32}, R_{14}, R_{24}, R_{34}] = [\ell, 0.94\ell, 0.88\ell, 0.82\ell, 0.76\ell, 0.70\ell]$, and (b) its corresponding orientation workspace in the subspace $\{\phi, \sigma\}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

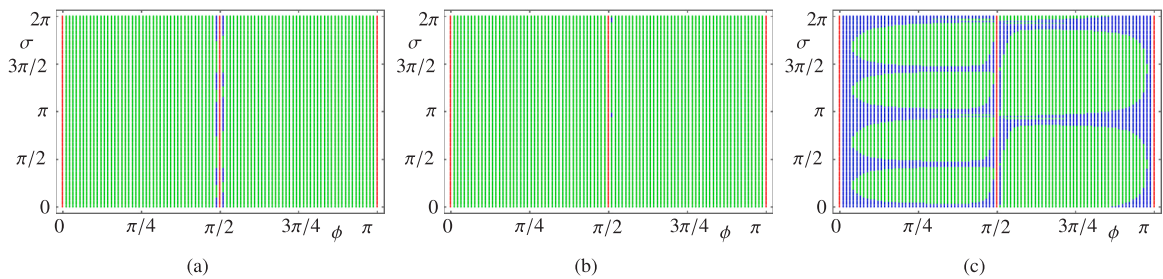


Fig. 10. The orientation workspaces of three SPMs from the studied robot class, with their geometries identical to those shown in Fig. 9(a), and their MP architectures specified as (a) $[\gamma_1, \gamma_2, \gamma_3] = [5\pi/9, 2\pi/3, 7\pi/9]$, (b) $[\gamma_1, \gamma_2, \gamma_3] = [11\pi/18, 11\pi/18, 7\pi/9]$, and (c) $[\gamma_1, \gamma_2, \gamma_3] = [4\pi/9, 4\pi/9, 10\pi/9]$, respectively.

while the remaining parameters are still assigned according to Eq. (60). This results in the design of the 3-RRR SPM depicted in Fig. 9(a), whose workspace is plotted in Fig. 9(b). As can be seen, this new set of geometric parameters almost completely eliminates the aforementioned mechanical interference of the robot, which significantly expands the range of tilt angle that allows for infinite torsional motion, ultimately spanning $\phi \in [2^\circ, 88^\circ]$ and $\phi \in [92^\circ, 178^\circ]$.

Lastly, it is worth noting that the large interference-free, singularity-free workspace is not only achievable for the specific design shown in Fig. 9(a) but can also be extended to a relatively wide range of designs within the studied robot class. To illustrate this, we continue using the specific link shapes shown in Fig. 9(a) and show that mechanical interference can be nearly completely eliminated by setting γ_2 — the angle between the distal-joint axes attached to the distal links with the largest and smallest central-axis radii (the red and yellow distal links in Fig. 9(a)) — to exceed 90° by some margin. For illustration, Fig. 10 presents the orientation workspaces of three designs from the studied robot class, with the first two complying the rule and the third not. As can be seen, mechanical interference is largely eliminated when the design rule is followed, with the remaining interference confined to the vicinity of singularities at $\phi = 90^\circ$, whereas violating the rule significantly restricts the workspace. Due to space limitations, the reasoning behind the design rule is omitted here, but can be found in the supplementary material for interested readers.

6.3. Workspace comparison and discussion

In this section, we conduct a comparative analysis of the workspace characteristics of the studied robot and three other SPMs with coaxial input axes. Specifically, these robots include the design example shown in Fig. 9(a), two 3-RRR SPMs⁷ studied by Bai et al. [43] and Tursynbek et al. [44], and a 3-(RSS)-S SPM studied by Enferadi et al. [42]. Moreover, this comparison focuses on a particular workspace characteristic of these robots, namely, the regions of MP normal vector directions that admit infinite torsional motion, as this represents a distinctive feature of SPMs with coaxial input axes. Furthermore, considering that mechanical interference is nearly completely eliminated in our design, and that interference heavily depends on link geometries, its effect has

⁷ Notice that the other two 3-RRR SPMs [43,44] used for comparison, although not belonging to the robot class under study, differ from our design only in their proximal-link dimensions, which are set to 60° and 45° , respectively, instead of 90° in our design.

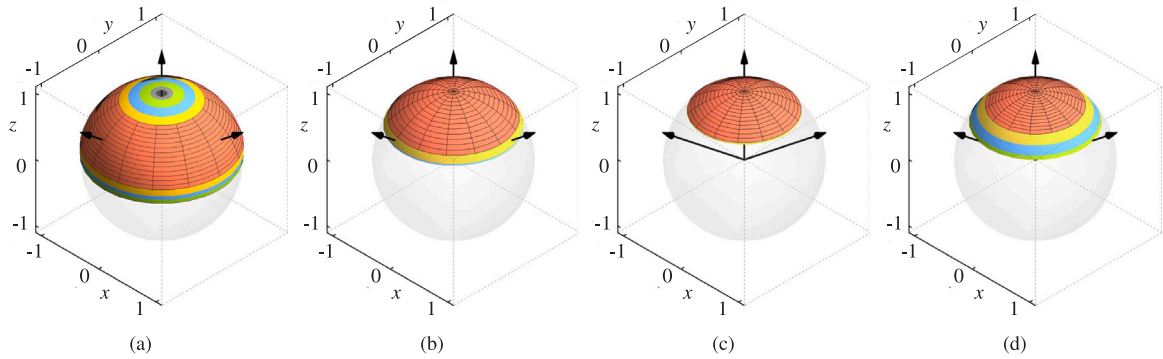


Fig. 11. Ranges of the MP normal vector directions that allow for infinite torsional motion under requirements of $LDI \geq 0.05$ (gray), 0.1 (green), 0.2 (blue), 0.3 (yellow), and 0.4 (orange) for (a) the 3-RRR SPM shown in Fig. 9(a), (b) the 3-RRR SPM studied by Bai et al. [43], (c) the 3-RRR SPM studied by Tursynbek et al. [44], and (d) the 3-(RSS)-S SPM studied by Enferadi et al. [42], where $[x, y, z] \equiv [\cos \psi \sin \phi, \sin \psi \sin \phi, \cos \phi]$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

Ranges of tilt angles $[\phi_{min}, \phi_{max}]$, the interval lengths $\Delta\phi$, and the areas of the ranges of the MP normal vector directions that admit infinite torsional motion under certain LDI thresholds (I. the 3-RRR SPM shown in Fig. 9(a), II. the 3-RRR SPM studied by Bai et al. [43], III. the 3-RRR SPM studied by Tursynbek et al. [44], and IV. the 3-(RSS)-S SPM studied by Enferadi et al. [42]).

	$[\phi_{min}, \phi_{max}]$				$\Delta\phi = \phi_{max} - \phi_{min}$				Area = $2\pi(\cos \phi_{min} - \cos \phi_{max})$			
	I	II	III	IV	I	II	III	IV	I	II	III	IV
$LDI \geq 0.05$	$[4^\circ, 88^\circ]$	$[0^\circ, 59^\circ]$	$[0^\circ, 44^\circ]$	$[0^\circ, 55^\circ]$	84°	59°	44°	55°	6.0486	3.0471	1.7634	2.6793
$LDI \geq 0.1$	$[8^\circ, 87^\circ]$	$[0^\circ, 59^\circ]$	$[0^\circ, 44^\circ]$	$[0^\circ, 55^\circ]$	79°	59°	44°	55°	5.8932	3.0471	1.7634	2.6793
$LDI \geq 0.2$	$[16^\circ, 84^\circ]$	$[0^\circ, 59^\circ]$	$[0^\circ, 44^\circ]$	$[0^\circ, 52^\circ]$	68°	59°	44°	52°	5.3830	3.0471	1.7634	2.4149
$LDI \geq 0.3$	$[24^\circ, 81^\circ]$	$[0^\circ, 58^\circ]$	$[0^\circ, 44^\circ]$	$[0^\circ, 45^\circ]$	57°	58°	44°	45°	4.7571	2.9536	1.7634	1.8403
$LDI \geq 0.4$	$[31^\circ, 76^\circ]$	$[0^\circ, 52^\circ]$	$[0^\circ, 43^\circ]$	$[0^\circ, 37^\circ]$	45°	52°	43°	37°	3.8657	2.4149	1.6880	1.2652

been neglected in the following analysis, although this may overestimate the workspace volumes of the three cited designs. Lastly, several LDI thresholds have been applied when evaluating the workspace to ensure a safe distance from singularities.

Next, the regions of MP normal vector directions that admit infinite torsional motion are evaluated for the aforementioned robots, as shown in Fig. 11. Moreover, the tilt-angle ranges $[\phi_{min}, \phi_{max}]$ of these regions, together with the corresponding interval lengths $\Delta\phi \equiv \phi_{max} - \phi_{min}$ and their areas, are all summarized in Table 2. As can be seen, the robot under study admits infinite torsional motion over a relatively large tilt-angle range when compared with the other three SPMs, particularly when the LDI threshold is relatively small. Moreover, even when assuming identical tilt-angle interval lengths for these robots, the “area” — representing the actual range of MP normal vector directions that allow infinite torsional motion — is significantly larger for our robot than for the other three, as shown in Fig. 11 and quantified in Table 2.

It is worth noting that the robot under study exhibits a singularity point at $\phi = 0^\circ$, which lies within the central region of the MP workspace—generally an undesirable location. Nevertheless, this singularity can be shifted away from the central region either by (a) selecting an alternative MP torsional axis that differs from the MP normal vector [48], or (b) redefining the reference posture so that the MP normal vector does not align with the coaxial input axes under the reference posture.

Besides using the aforementioned strategies to shift the singularity point, the proposed robot can also be applied in scenarios where the central workspace region is not critical. For example, cube-shaped, polyhedral, or spherical robots often need to generate flywheel-induced rotations about three orthogonal directions to achieve forward/backward locomotion, lateral motion, and in-place turning [59,60]. This can be accomplished either by employing three orthogonally arranged flywheels or by using a single flywheel whose axis can be reoriented relative to the robot body along three orthogonal directions. In the latter case, the flywheel is first spun up and then rapidly decelerated to induce a rolling or flipping motion about the corresponding rotation axis [59]. In such scenarios, the three orthogonal directions indicated by the black arrows in Fig. 11(a) can serve as the three rotation axes for unlimited torsional motion, all lying within regions of good dexterity, making our design suitable for these tasks. On the other hand, most existing 3-RRR SPMs either fail to cover three orthogonal axes admitting unlimited torsional motion or exhibit poor kinematic performance in these directions, as shown in Fig. 11(b)–(d).

For further details on these strategies and potential applications of the studied robot class in light of the aforementioned singularity point, see the supplementary material. Additionally, beyond the workspace comparison, the singularity loci and LDI distributions of the aforementioned robots are also provided to offer a comprehensive assessment of their singularity-free workspaces and kinematic performance, which are also available in the supplementary material for interested readers.

7. Conclusions

A special class of 3-RRR SPMs was investigated in this paper, characterized by coaxial input-joint axes, coplanar distal-joint axes, and six orthogonal links. Two novel formulations were proposed for the FK of the robot class, both yielding linear univariate polynomials that significantly streamline the FK formulation and reduce the computational cost. Moreover, it was found that all instances of the robot class share identical singularity loci, which are both intuitive and physically interpretable; specifically, singularities arise iff the MP tilt angle ϕ equals $k\pi/2$, with $k = 0, 1, 2$, demonstrating the potential for the robot class to produce a large singularity-free orientation workspace; additionally, the corresponding singularity conditions are also derived in the input joint space, thereby facilitating singularity avoidance and path planning. Furthermore, the local and global dexterity indices were evaluated for the robot class under different MP architectural parameters, revealing that two MP architectures, including a symmetric one, both achieve optimal dexterity performance. Notably, the proposed FK formulations, along with the special characteristics in terms of singularity and dexterity distributions, can also be applied to a variety of SPMs, provided that they are kinematically equivalent to 3-RRR SPMs and that their geometries are tuned such that their 3-RRR kinematic equivalents fall within the robot class under study. Finally, the workspace analysis shows that the robot class demonstrates a considerably large tilt-torsion orientation workspace when the architectural parameters and link shapes are appropriately specified, allowing for infinite torsional motion when pointed within almost a hemisphere, outperforming most previously reported SPMs in this regard [42–44]. In summary, the above interesting features make the robot class under study a promising candidate for various applications, such as orientation devices, robotic wrists and shoulders [1,2].

CRedit authorship contribution statement

Shuai Zhang: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Wei Li:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Huayan Pu:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Rui Chen:** Visualization, Methodology, Investigation, Formal analysis. **Jun He:** Validation, Methodology, Investigation, Formal analysis. **Jun Luo:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.mechmachtheory.2025.106288>.

Data availability

No data was used for the research described in the article.

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